



Brief paper

Linear-in-parameters neural adaptive observers for nonlinear systems in observability canonical form[☆]

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ABSTRACT

Nonlinear adaptive observers that do not require knowledge of the dynamics of the system being observed are proposed. This objective is pursued by envisioning a linear-in-parameters adaptive law that allows one to estimate the system dynamics, which are assumed to be in observability canonical form, just from measurements of the output. In the case that a parametric model for the plant is known, the proposed adaptive observer asymptotically reconstructs both the state and the unknown parameters of the observed system. On the other hand, if a parametric model for the system is not available, then the reconstruction of both its state and of its model is practical. The effectiveness of these observers is validated by numerical simulations.

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1. Introduction

The problem of reconstructing the state of a dynamical system is of main importance in automatic control. In fact, it plays a crucial role in monitoring the system (Huh & Stein, 1994), detecting faults (Selmic et al., 2009), and achieving good performance in the presence of disturbances (Song & Lewis, 2020). Classical observers for nonlinear plants require exact knowledge of the dynamics of the system being observed (Bonnabel & Slotine, 2014; Ciccarella et al., 1993). However, for several practical processes, defining an exact model is an extremely challenging task. A large research effort has then been spent to design observers that do not require exact knowledge of the dynamics of the system being observed. In Sections 5.3, 5.4, and 5.5 of Ioannou and Sun (2012), it has been shown how to design adaptive observer for linear systems in observability form. By exploiting the universal approximation property of neural networks (Cybenko, 1989), several attempts have been made to design adaptive observers also for nonlinear systems: in Ahmed and Riyaz (2000), neural networks are used to approximate the nonlinear Kalman gain

of the observer; in Kim et al. (1996), the state of affine single-input-single-output nonlinear systems are estimated by using linear-in-parameters neural network models (i.e., the tuned parameters appear linearly in the model); nonlinear-in-parameters neural network models have been used in Abdollahi et al. (2006) and Hovakimyan et al. (2002) to estimate the state of a nonlinear system; in Adhyaru (2012), observers for non-linear systems are designed by approximating the solution to an optimal control problem via a neural network; in Lakhal et al. (2010) and Vargas and Hemerly (2000), weights adaptation strategies have been given to adapt the weight of a neural network modeling the unknown non-linearities of the plant; in Abdollahi et al. (2006), Luenberger-like neural observers have been used to estimate the state of flexible-joint manipulators; in Leu et al. (1999), an observer-based adaptive fuzzy-neural controller for a class of unknown non-linear systems has been proposed; whereas, in Chakrabarty et al. (2019), a data-driven adaptive observer has been designed using neural approximators with Lipschitz activations. Differently from the above mentioned references, wherein different paradigms (such as extended Kalman filtering, Luenberger-like observers, or backpropagation-based methods) have been used to design neural adaptive observers for nonlinear plants, in this paper, high-gain observers are used to design adaptive laws able of reconstructing the state and of providing a model for the dynamics of the system.

High-gain observers have been already used in the literature to design adaptive observers for nonlinear systems: in Bin and Marconi (2021) an extended high-gain observer is combined with a discrete-time identifier to design an adaptive observer

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for nonlinear systems, whereas in [Buisson-Fenet et al. \(2021\)](#) high-gain observers are coupled with a non-parametric Gaussian process model to jointly estimate the state and the dynamics of a nonlinear plant. Differently from the above mentioned techniques, the adaptive high-gain observers proposed in this paper continuously adapt the estimated parameters without resorting to discrete-time computations and hence possibly reducing the computational burden.

2. Observability of nonlinear systems

The notation used in this paper is the same as in [Ioannou and Sun \(2012\)](#). Consider the nonlinear system

$$\dot{z}(t) = f(z(t)), \quad y(t) = h(z(t)), \quad (1)$$

where $z(t) \in \mathbb{R}^p$ is the state, $y(t) \in \mathbb{R}$ is the output, f and h are real analytic functions on the whole $\mathbb{R}^p$. Let $L_f^0 h(x) := h(x)$ and $L_f^i h(x) := (\frac{\partial}{\partial z} L_f^{i-1} h(z))f(z)$ for $i \in \mathbb{Z}$, $i > 1$. The map $z \mapsto O_n(z)$, $O_n(z) = \text{col}(L_f^0 h(z), \dots, L_f^{n-1} h(z))$, is the *observability map of order n of system (1)* that relates the time derivatives of the output y (up to order $n - 1$) with its state z . If O_n is injective, then there is a *left inverse* O_n^{-1} such that $O_n^{-1} \circ O_n(z) = z$, $\forall z \in \mathbb{R}^p$, and system (1) is *n -differentially observable*. If system (1) is *n -differentially observable*, then it can be immersed into the *observability canonical form* ([Gauthier & Kupka, 2001](#)):

$$\dot{x}_i(t) = x_{i+1}(t), \quad i = 1, \dots, n-1 \quad (2a)$$

$$\dot{x}_n(t) = v(x(t)), \quad (2b)$$

$$y(t) = x_1(t). \quad (2c)$$

where $x(t) = O_n(z(t))$ and $v(x) = L_f^n h \circ O_n^{-1}(x)$. Assuming that system (1) has been already recast into the form (2), the main objective of this work is to design adaptive observers for system (2) that do not require the knowledge of the function v governing the dynamics (2). Such a goal has already been partially pursued in [Tornambe \(1992\)](#) by using the *high-gain observer*

$$\dot{\hat{x}}_p(t) = A_\varepsilon \hat{x}_p(t) + B_\varepsilon y(t), \quad (3)$$

where $\hat{x}_p(t) \in \mathbb{R}^n$ is an estimate of $x(t)$, the matrices A_ε and B_ε are given by

$$A_\varepsilon := \begin{bmatrix} -\frac{k_1}{\varepsilon} & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{k_{n-1}}{\varepsilon^{n-1}} & 0 & \dots & 1 \\ -\frac{k_n}{\varepsilon^n} & 0 & \dots & 0 \end{bmatrix}, \quad B_\varepsilon := \begin{bmatrix} \frac{k_1}{\varepsilon} \\ \vdots \\ \frac{k_{n-1}}{\varepsilon^{n-1}} \\ \frac{k_n}{\varepsilon^n} \end{bmatrix}, \quad (4)$$

$k_1, \dots, k_n$ are positive constants such that $s^n + k_1 s^{n-1} + \dots + k_{n-1} s + k_n$ is Hurwitz, and $\varepsilon > 0$ is a tuning parameter (the scalar $\frac{1}{\varepsilon}$ is usually referred to as the *observer gain*). In particular, if there exists $\Gamma > 0$ such that $|v(x(t))| \leq \Gamma$ for all $t \geq 0$, the norm of the estimation error $\tilde{x}_p := \hat{x}_p - x_p$ can be made arbitrarily small in an arbitrarily small amount of time by letting the parameter ε be sufficiently small.

It is worth pointing out that if the dynamics of system (1) are unknown, the state z in the representation (1) cannot be determined, in general, from just output measurements. In fact, given any diffeomorphism $\bar{z} = \mathcal{E}(z)$, with inverse $z = \mathcal{E}^{-1}(\bar{z})$, the output y of system (1) is indistinguishable from that of $\dot{\bar{z}} = \frac{\partial \mathcal{E}}{\partial z}(\mathcal{E}^{-1}(\bar{z}))f(\mathcal{E}^{-1}(\bar{z}))$, $y = h(\mathcal{E}^{-1}(\bar{z}))$, with $\bar{z}(0) = \mathcal{E}(z(0))$. If, instead, the dynamics (1) are known, then the state z in the representation (2) can be estimated as $\hat{z} = O_n^{-1}(\hat{x})$, possibly using a dynamical inversion algorithm ([Menini et al., 2017b](#)), where $\hat{x}$ is an estimate of the state x of system (2).

Although the high-gain observer (3) practically estimates the state of system (2), it presents two drawbacks: the estimation

error need not converge asymptotically to zero, and the observer gain $\frac{1}{\varepsilon}$ has to be quite large to ensure that the practical estimate $\hat{x}_p(t)$ is eventually close to the actual state $x(t)$ of system (2). The main objective of this work is to mitigate these two issues by envisioning an adaptation strategy that allows one to estimate the unknown function v by just using measurements of the output y . This objective is pursued in Section 3 assuming that the function v is known in parametric form. Such a hypothesis is then removed in Section 4 by using neural networks to approximate v .

3. Adaptive observers for nonlinear systems

The following assumption is made all throughout this section to design adaptive observers for system (2).

Assumption 1. The function v in (2b) is known in parametric form, i.e., there exist a known feature vector $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and an unknown coefficient vector $\lambda \in \mathbb{R}^m$ such that $v(x) = \lambda^\top \phi(x)$.

Assumption 1 essentially states that the function v governing the dynamics of system (2) is known in parametric form. It is worth pointing out that if the dynamics of system (1) are known in parametric form and such a system is *n -differentially observable* for any selection of the parameters, then it can be immersed into the form (2) with v satisfying Assumption 1. Under this assumption, let $H := \text{col}(0, \dots, 0, 1)$, $C_0 := [1 \ 0 \ \dots \ 0]$, and consider the observer with state $(\hat{x}_a(t), \hat{\lambda}_a(t)) \in \mathbb{R}^n \times \mathbb{R}^m$,

$$\dot{\hat{x}}_a(t) = A_\varepsilon \hat{x}_a(t) + B_\varepsilon y(t) + H \hat{\lambda}_a^\top(t) \phi(\hat{x}_a(t)), \quad (5a)$$

$$\dot{\hat{\lambda}}_a(t) = -\gamma (\hat{y}_a(t) - y(t)) \phi(\hat{x}_a(t)), \quad (5b)$$

where $\hat{y}_a = C_0 \hat{x}_a$, $\gamma > 0$ is the *adaptation gain* and the matrices A_ε and B_ε are as in (4).

Lemma 1. Let Assumption 1 hold, let ϕ be bounded and globally Lipschitz, and consider system (5). There exist constants $\varepsilon^* > 0$ and $\hat{\Theta}^* > 0$ such that, if $\varepsilon \leq \varepsilon^*$ and the signal $\hat{\phi}_{\varepsilon,a}(t) := ((sI - \frac{1}{\varepsilon} A_1)^{-1} H \phi^\top(\hat{x}_a(t)))^\top$ satisfies

$$\check{c}_1 \varepsilon^2 I \leq \int_{t_0}^{t_0 + \hat{T}_{\varepsilon,0}} \hat{\phi}_{\varepsilon,a}(\tau) \hat{\phi}_{\varepsilon,a}^\top(\tau) d\tau \leq \check{c}_2 \varepsilon^2 I, \quad (6)$$

for all $t_0 \geq 0$, where $\check{c}_1 > 0$ and $\check{c}_2 > 0$ are independent of ε and $\frac{\hat{T}_{\varepsilon,0}}{\varepsilon^1} \leq \hat{\Theta}^*$ for all $\varepsilon \in (0, \varepsilon^*)$, then one has $\lim_{t \rightarrow +\infty} \hat{x}_a(t) - x(t) = 0$ and $\lim_{t \rightarrow +\infty} \hat{\lambda}_a(t) - \lambda = 0$.

Proof. Since ϕ is globally Lipschitz, there exists $L > 0$ such that $\|\phi(\hat{x}) - \phi(x)\| \leq L \|\hat{x} - x\|$, $\forall \hat{x}, x \in \mathbb{R}^n$. Furthermore, by the multivariate version of the Taylor theorem ([Königsberger, 2013](#)), there exists a function $R(x, \hat{x})$, $\|R(x, \hat{x})\| \leq L$, $\forall x, \hat{x} \in \mathbb{R}^n$, such that $\phi(\hat{x}) - \phi(x) = R(x, \hat{x})\tilde{x}$. Define the errors $\tilde{x}_a := \hat{x}_a - x$, $\tilde{x}_{\varepsilon,a} := E_\varepsilon \tilde{x}_a$, where

$$E_\varepsilon := \text{diag}(1, \varepsilon, \dots, \varepsilon^{n-1}), \quad (7)$$

$\tilde{\lambda}_a := \hat{\lambda}_a - \lambda$, and $\tilde{\xi}_{\varepsilon,a} := \text{col}(\tilde{x}_{\varepsilon,a}, \tilde{\lambda}_a)$, whose dynamics are given by $\dot{\tilde{\xi}}_{\varepsilon,a}(t) = \bar{A}_a(t) \tilde{\xi}_{\varepsilon,a}(t)$, with

$$\bar{A}_a(t) := \begin{bmatrix} \frac{1}{\varepsilon} A_1 + E_\varepsilon H \lambda^\top R(x(t), \hat{x}_a(t)) E_\varepsilon^{-1} & E_\varepsilon H \phi^\top(\hat{x}_a(t)) \\ -\gamma \phi(\hat{x}_a(t)) C_0 & 0 \end{bmatrix}.$$

By Lemma 5 of [Carnevale et al. \(2018\)](#), the transfer function $W_0(s) := C_0 (sI - A_1)^{-1} H$ equals $\frac{1}{s^n + k_1 s^{n-1} + \dots + k_{n-1} s + k_n}$. Thus, letting $d := \frac{2}{k_n}$, the function $W_d(s) := d + C_0 (sI - A_1)^{-1} H$ is strictly positive real by Theorem 3.5.1 of [Ioannou and Sun \(2012\)](#). Thus, since the matrix A_1 is stable and the pair (A_1, H) is controllable,

by Lemma 3.5.3 of [Ioannou and Sun \(2012\)](#), there exist a positive definite matrix $P_a \in \mathbb{R}^{n \times n}$, a vector $q_a \in \mathbb{R}^n$, and $\nu_a > 0$ such that

$$A_1^\top P_a + P_a A_1 = -q_a q_a^\top - \nu_a I, \quad (8a)$$

$$P_a H - C_0^\top = \sqrt{2d} q_a. \quad (8b)$$

Thus, consider the positive definite function

$$V_a(t) := \tilde{x}_{\varepsilon,a}^\top(t) P_a \tilde{x}_{\varepsilon,a}(t) + \frac{\varepsilon^{n-1}}{\gamma} \tilde{\lambda}_a^\top(t) \tilde{\lambda}_a(t). \quad (9)$$

Since $E_\varepsilon H = \varepsilon^{n-1} H$ and $C_0 E_\varepsilon = C_0$, using the completing the squares procedure (Algorithm 1 of [Menini et al., 2017a](#)), the time derivative of V_a satisfies

$$\begin{aligned} \dot{V}_a(t) \leq & - \left(\frac{\nu_a}{\varepsilon} - 2L \|P_a\| \|\lambda\| \right) \tilde{x}_{\varepsilon,a}^\top(t) \tilde{x}_{\varepsilon,a}(t) \\ & - \frac{1}{\varepsilon} (q_a^\top \tilde{x}_{\varepsilon,a}(t) - \sqrt{2d} \varepsilon^{n-1} \tilde{\lambda}_a^\top(t) \phi(\hat{x}_a(t)))^2 \\ & + 2d \varepsilon^{2n-1} (\tilde{\lambda}_a^\top(t) \phi(\hat{x}_a(t)))^2. \end{aligned} \quad (10)$$

Consider now the output matrix $\bar{C}_a := [I \ 0]$ and the output feedback gain

$$K_a(t) := \begin{bmatrix} \varepsilon^{n-1} H \lambda^\top R(x(t), \hat{x}_a(t)) E_\varepsilon^{-1} \\ -\gamma \phi(\hat{x}_a(t)) C_0 \end{bmatrix},$$

whose norm is bounded by a constant $\bar{K}_a$ that is independent of ε , since ϕ is bounded, $R(x, \hat{x}) \leq L$ for all $x, \hat{x} \in \mathbb{R}^n$, and $\varepsilon^{n-1} \|E_\varepsilon^{-1}\| \leq 1$. Hence, define

$$\begin{aligned} \check{A}_a(t) &:= \bar{A}_a(t) + K_a(t) \bar{C}_a \\ &= \begin{bmatrix} \frac{1}{\varepsilon} A_1 & \varepsilon^{n-1} H \phi^\top(\hat{x}_a(t)) \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Since the signal $\hat{\phi}_{\varepsilon,a}(t)$ satisfies (6), the signal $\hat{\phi}_{\varepsilon,a,s}(t) := \varepsilon^{n-1} ((sI - \frac{1}{\varepsilon} A_1)^{-1} H \phi^\top(\hat{x}_a(t)))^\top$ satisfies $\check{c}_1 \varepsilon^{2n} I \leq \int_{t_0}^{t_0 + \hat{T}_{\varepsilon,0}} \hat{\phi}_{\varepsilon,a,s}^\top(\tau) \hat{\phi}_{\varepsilon,a,s}(\tau) d\tau \leq \check{c}_2 \varepsilon^{2n} I$, for all $t_0 \geq 0$. By Lemma 4.8.4 of [Ioannou and Sun \(2012\)](#), this implies that the pair $(\check{A}_a(t), \bar{C}_a)$ is uniformly completely observable. In particular, letting $\check{\Phi}_a(t, t_0)$ be the state transition matrix of $\check{A}_a(t)$, in the limiting case $\varepsilon \rightarrow 0$, one has

$$\text{that } \varepsilon^{n+1} I \leq \int_{t_0}^{t_0 + \frac{4\hat{T}_{\varepsilon,0}}{\varepsilon^{n-1}\check{c}_1}} \check{\Phi}_a^\top(\tau, t_0) \bar{C}_a^\top \bar{C}_a \check{\Phi}_a(\tau, t_0) d\tau \leq 8 \frac{\check{c}_2}{\check{c}_1} \varepsilon^{n+1} I,$$

for all $t_0 \geq 0$. Therefore, by Lemma 4.8.1 of [Ioannou and Sun \(2012\)](#), the pair $(\check{A}_a(t), \bar{C}_a)$ is uniformly completely observable. Namely, letting $\bar{\Phi}_a(t, t_0)$ be the state transition matrix of $\bar{A}_a(t)$, in the limiting case $\varepsilon \rightarrow 0$, one has that, for all $t_0 \geq 0$,

$$\begin{aligned} & \frac{\varepsilon^{n+1}}{2(1 + 32\varepsilon^2 \frac{\check{c}_2}{\check{c}_1} \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1} \bar{K}_a^2)} I \\ & \leq \int_{t_0}^{t_0 + \frac{4}{\varepsilon^{n-1}} \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1}} \bar{\Phi}_a^\top(\tau, t_0) \bar{C}_a^\top \bar{C}_a \bar{\Phi}_a(\tau, t_0) d\tau. \end{aligned} \quad (11)$$

Consider now the output matrix $\bar{D}_a(t) := [0 \ \phi(\hat{x}_a(t))]$ and the output feedback gain $J_a := \text{col}(\varepsilon^{n-1} H, 0)$. Thus, define the time-varying matrix

$$\begin{aligned} \check{A}_a(t) &:= \bar{A}_a(t) + J_a(t) \bar{D}_a(t) \\ &= \begin{bmatrix} \frac{1}{\varepsilon} A_1 + E_\varepsilon H \lambda^\top R(x(t), \hat{x}_a(t)) E_\varepsilon^{-1} & 0 \\ -\gamma \phi(\hat{x}_a(t)) C_0 & 0 \end{bmatrix}, \end{aligned}$$

whose associated linear, time-varying system is $\dot{z}_1(t) = (\frac{1}{\varepsilon} A_1 + \varepsilon^{n-1} H \lambda^\top R(x(t), \hat{x}_a(t)) E_\varepsilon^{-1}) z_1(t)$, $\dot{z}_2(t) = -\gamma \phi(\hat{x}_a(t)) C_0 z_1(t)$. By Section 14.5 of [Khalil \(2002\)](#), in the limiting case $\varepsilon \rightarrow 0$, there exist $\alpha_1 > 0$ and $\alpha_2 > 0$, independent of ε , such that $\|z_1(t)\| \leq \alpha_1 e^{-\frac{\alpha_2}{\varepsilon}(t-t_0)} \|z_1(t_0)\|$. Therefore, it results that $\|z_2(t)\| \leq \|z_2(t_0)\| + \frac{\alpha_1}{\alpha_2} \varepsilon \gamma \phi_a \|z_1(t_0)\|$, for all $t \geq t_0$, where

$\bar{\phi}_a = \sup_{\tau \geq 0} \|\phi(\hat{x}_a(\tau))\|$. Thus, by Lemma 4.8.1 of [Ioannou and Sun \(2012\)](#), in the limiting case $\varepsilon \rightarrow 0$, there exists a constant $\alpha_3 > 0$, independent of ε , such that, for all $t_0 \geq 0$ and all $T \geq 0$,

$$\begin{aligned} & \int_{t_0}^{t_0+T} \bar{\Phi}_a^\top(\tau, t_0) \bar{D}_a^\top(\tau) \bar{D}_a(\tau) \bar{\Phi}_a(\tau, t_0) d\tau \\ & \leq \alpha_3 T e^{\varepsilon^{2n-2} \alpha_3 T^2}. \end{aligned} \quad (12)$$

By (10), (11), and (12), one has that, for all $t_0 \geq 0$, $V_a(t_0 + \frac{4}{\varepsilon^{n-1}} \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1}) - V_a(t_0) \leq -\alpha_\varepsilon \|\tilde{x}_{\varepsilon,a}(t_0)\|^2$, $\alpha_\varepsilon := \varepsilon^n \frac{\nu_a - 2\varepsilon L \|P_a\| \|\lambda\|}{2(1 + 32\varepsilon^2 \frac{\check{c}_2}{\check{c}_1} \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1} \bar{K}_a^2)} -$

$8\varepsilon^n d \alpha_3 \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1} e^{16\alpha_3 \frac{\hat{T}_{\varepsilon,0}^2}{\check{c}_1^2}}$. Thus, there exist $\varepsilon^* > 0$ and $\Theta^* > 0$ such

that, if $\varepsilon \leq \varepsilon^*$ and $\frac{\hat{T}_{\varepsilon,0}}{\check{c}_1} \leq \Theta^*$ for all $\varepsilon \in (0, \varepsilon^*)$, then

$$\begin{aligned} V_a\left(t_0 + \frac{4}{\varepsilon^{n-1}} \frac{\hat{T}_{\varepsilon,0}}{\check{c}_1}\right) & \leq V_a(t_0) - \varepsilon^n \frac{\nu_a}{8} \|\tilde{x}_{\varepsilon,a}(t_0)\|^2 \\ & \leq \left(1 - \frac{\varepsilon^n \nu_a}{8 \max\{\frac{\varepsilon^{n-1}}{\gamma}, \bar{\sigma}\}}\right) V_a(t_0), \end{aligned}$$

where $\bar{\sigma}$ denotes the maximum eigenvalue of P_a . Therefore, the statement follows by the same reasoning used to prove Theorem 1.5.2 of [Sastri and Bodson \(2011\)](#). $\square$

Note that system (5) just requires measurements of the output y of system (2) in order to be implemented. However, in the case that $\hat{\phi}_{\varepsilon,a}(t)$ does not satisfy the persistence of excitation condition (6), the estimate $\hat{\lambda}_a$ may drift to infinity with time. In fact, by (10), the time derivative of the function V_a defined in (9) is not sign definite and hence ultimate boundedness of the trajectories of system (5) cannot be guaranteed. This phenomenon is well known in the adaptive control literature as *parameter drift* ([Ioannou & Kokotovic, 1983](#)). In order to avoid this undesired effect, consider the following adaptive observer with state $(\hat{x}_b(t), \hat{\lambda}_b(t)) \in \mathbb{R}^n \times \mathbb{R}^m$,

$$\dot{\hat{x}}_b(t) = A_\varepsilon \hat{x}_b(t) + B_\varepsilon y(t) + H \hat{\lambda}_b^\top(t) \phi(\hat{x}_b(t)), \quad (13a)$$

$$\dot{\hat{\lambda}}_b(t) = \mathcal{P}_M (-\gamma \tilde{y}_b(t) \phi(\hat{x}_b(t))), \quad (13b)$$

where $\tilde{y}_b := \hat{y}_b - y$, $\hat{y}_b := C_0 \hat{x}_b$, $\mathcal{P}_M$ denotes the *projection operator*,

$$\mathcal{P}_M(-\gamma \tilde{y} \phi) := \begin{cases} -\gamma \tilde{y} \phi, & \text{if } \{\|\hat{\lambda}\|^2 < M\} \text{ or} \\ & \{\|\hat{\lambda}\|^2 = M \text{ and } \tilde{y} \hat{\lambda}^\top \phi \geq 0\}, \\ 0, & \text{otherwise,} \end{cases}$$

and $M > 0$ is a sufficiently large parameter such that $\|\lambda\|^2 \leq M$. By (13b), if $\|\hat{\lambda}_b(0)\|^2 \leq M$, then $\|\hat{\lambda}_b(t)\|^2 \leq M$ for all $t \geq 0$, i.e., the set $\{\hat{x}_b \times \hat{\lambda}_b \in \mathbb{R}^n \times \mathbb{R}^m : \|\hat{\lambda}_b\|^2 \leq M\}$ is positively invariant for system (13).

Theorem 1. Let [Assumption 1](#) hold, suppose that ϕ is globally Lipschitz and that the solution $x(t)$ of system (2) is bounded. Consider system (13) and let $\|\hat{\lambda}_b(0)\|^2 \leq M$. For each $\varrho > 0$, $\Delta > 0$, and $T > 0$, there is $\varepsilon^* > 0$ such that, if $\varepsilon < \varepsilon^*$ and $\|\hat{x}_b(0) - x(0)\|^2 \leq \Delta$, then $\hat{x}_b(t)$ and $\hat{\lambda}_b(t)$ are bounded and $\|\hat{x}_b(t) - x(t)\|^2 \leq \varrho$, $\forall t \geq T$.

Proof. Since the matrix A_1 is stable, there exists a symmetric, positive definite matrix P solving

$$A_1^\top P + P A_1 = -I. \quad (14)$$

Thus, defining $\tilde{x}_b := \hat{x}_b - x$, $\tilde{x}_{\varepsilon,b} := E_\varepsilon \tilde{x}_b$ where E_ε is as in (7), and $\tilde{\lambda}_b := \hat{\lambda}_b - \lambda$, define $q_b := P H$ and $V_b(t) = \tilde{x}_{\varepsilon,b}^\top(t) P \tilde{x}_{\varepsilon,b}(t)$, whose time derivative satisfies

$$\begin{aligned} \dot{V}_b(t) &= -\frac{1}{\varepsilon} \|\tilde{x}_{\varepsilon,b}(t)\|^2 + 2\varepsilon^{n-1} \tilde{x}_{\varepsilon,b}^\top(t) q_b \tilde{\lambda}_b^\top(t) \phi(\hat{x}_b(t)) \\ & \quad + 2\varepsilon^{n-1} \tilde{x}_{\varepsilon,b}^\top(t) q_b \lambda^\top (\phi(\hat{x}_b(t)) - \phi(x(t))). \end{aligned}$$

By completing the squares and using the fact that ϕ is globally Lipschitz, one has that

$$\dot{V}_b(t) \leq -\left(\frac{1}{\varepsilon} - 2L\|q_b\|\|\lambda\| - \|q_b\|^2\right)\|\tilde{x}_{\varepsilon,b}(t)\|^2 + \varepsilon^{2n-2}(\tilde{\lambda}_b^\top(t)\phi(\hat{x}_b(t)))^2.$$

Thus, by considering that there exists $\bar{M} > 0$ such that $\|\tilde{\lambda}_b(t)\|^2 \leq \bar{M}$ for all $t \geq 0$, since $\|\hat{\lambda}_b(t)\|^2 \leq M$, and that $\|\phi(\hat{x}_b(t))\| \leq \|\phi(x(t))\| + \frac{1}{\varepsilon^{n-1}}L\|\tilde{x}_{\varepsilon,b}(t)\|$, letting $\bar{\phi} := \sup_{t \geq 0} \phi(x(t))$, which is a bounded constant since $x(t)$ is bounded, and using the inequality $(a+b)^2 \leq 2a^2 + 2b^2$, one has that $\dot{V}_b(t) \leq 2\varepsilon^{2n-2}\bar{M}\bar{\phi}^2 - \left(\frac{1}{\varepsilon} - 2L\|q_b\|\|\lambda\| - \|q_b\|^2 - 2\bar{M}L^2\right)\|\tilde{x}_{\varepsilon,b}(t)\|^2$. Therefore, there exists $\varepsilon^\diamond > 0$ such that, if $\varepsilon < \varepsilon^\diamond$ and $\|\tilde{x}_{\varepsilon,b}\|^2 >$

$\frac{2\varepsilon^{2n-1}\bar{M}\bar{\phi}^2}{1-2\varepsilon L\|q_b\|\|\lambda\| - \varepsilon\|q_b\|^2 - 2\varepsilon\bar{M}L^2}$, then $\dot{V} < 0$, which implies, by Theorem 4.18 of Khalil (2002), that there exist constants $\varpi_1 > 0$ and $\varpi_2 > 0$, independent of ε , such that, for all $t \geq t_0$, 0,

$$\|\tilde{x}_{\varepsilon,b}(t)\|^2 \leq \varpi_1 e^{-\frac{t-t_0}{2\varepsilon}}\|\tilde{x}_{\varepsilon,b}(t_0)\|^2 + \varepsilon^{2n-1}\varpi_2. \quad (15)$$

Note that $T_\varepsilon := -2\varepsilon \log\left(\frac{\varepsilon^{2n-1}\varpi_2}{\varpi_1\|\tilde{x}_{\varepsilon,b}(t_0)\|^2}\right) \leq 2\varepsilon \log\left(\frac{\varpi_1\Delta}{\varpi_2\varepsilon}\right)$ is such that $\varpi_1 \exp\left(-\frac{1}{2\varepsilon}(t-t_0)\right)\|\tilde{x}_{\varepsilon,b}(t_0)\|^2 \leq \varepsilon^{2n-1}\varpi_2$, for all $t \geq t_0 + T_\varepsilon$. Thus, the statement follows by the fact that $\|\tilde{x}_b(t)\|^2 \leq \frac{1}{\varepsilon^{2n-2}}\|\tilde{x}_{\varepsilon,b}(t)\|^2$, $\forall t \geq 0$, and that $\lim_{\varepsilon \rightarrow 0} T_\varepsilon = 0$. $\square$

Lemma 2. Suppose that there exists $\bar{\varepsilon} > 0$ such that the signal $\varphi_\varepsilon(t) := ((sI - A_\varepsilon)^{-1}H\phi^\top(x(t)))^\top$ satisfies

$$c_1\varepsilon^2I \leq \int_{t_0}^{t_0+T_{\varepsilon,0}} \varphi_\varepsilon(\tau)\varphi_\varepsilon^\top(\tau)d\tau \leq c_2\varepsilon^2I, \quad (16)$$

for all times $t_0 \geq 0$, for some constants $c_1 > 0$ and $c_2 > 0$ independent of ε , and for all $\varepsilon \in (0, \bar{\varepsilon})$. Then, for each $T > 0$ and $\Delta > 0$, there exists $\varepsilon^* \in (0, \bar{\varepsilon})$ such that, if $\|\tilde{x}_b(0)\| \leq \Delta$, then, for all $\varepsilon \in (0, \varepsilon^*)$, the signal $\hat{\varphi}_{\varepsilon,b}(t) := ((sI - A_\varepsilon)^{-1}H\phi^\top(\hat{x}_b(t)))^\top$ satisfies

$$\check{c}_1\varepsilon^2I \leq \int_{t_0}^{t_0+\hat{T}_{\varepsilon,0}} \hat{\varphi}_{\varepsilon,b}(\tau)\hat{\varphi}_{\varepsilon,b}^\top(\tau)d\tau \leq \check{c}_2\varepsilon^2I, \quad (17)$$

for all $t_0 \geq T$ and some $\check{c}_1 > 0$, $\check{c}_2 > 0$ independent of ε .

Proof. In order to establish that $\hat{\varphi}_{\varepsilon,b}(t)$ is persistently exciting, it suffices to find $\hat{\beta}_{\varepsilon,1} > 0$, $\hat{\beta}_{\varepsilon,2} > 0$, and $\hat{T}_{\varepsilon,0} > 0$ such that $\hat{\beta}_{\varepsilon,1} \leq \int_{t_0}^{t_0+\hat{T}_{\varepsilon,0}} \|q^\top \hat{\varphi}_{\varepsilon,b}(\tau)\|^2 d\tau \leq \hat{\beta}_{\varepsilon,2}$, for all $t_0 \geq T$ and all $q \in \mathbb{R}^m$ such that $\|q\| = 1$.

First, note that $\phi(\hat{x}_b(t)) = \phi(x(t)) + \theta_b(t)$, with $\theta_b(t) := \phi(\hat{x}_b(t)) - \phi(x(t))$, $\|\theta_b(t)\| \leq L\|\tilde{x}_b(t)\|$. Secondly, by the same reasoning used to prove Theorem 1, the system $\dot{\chi}(t) = A_\varepsilon \chi(t) + H\theta_b(t)$, is input-to-state stable with input-state gain $\varepsilon\varpi_3$, where ϖ_3 is a constant independent of ε . Therefore, by Theorem 1 and (15), letting $\vartheta_{\varepsilon,b}(t) := ((sI - A_\varepsilon)^{-1}H\theta_b(t))^\top$, one has that there exists $\varepsilon^\diamond > 0$ such that, if $\varepsilon \in (0, \varepsilon^\diamond)$, then $\|\vartheta_{\varepsilon,b}(t)\|^2 \leq \varepsilon^3\varpi_4$ for all $t \geq T$, where ϖ_4 is a constant independent of ε .

Thus, since $(a+b)^2 \leq 2a^2 + 2b^2$, one has that, $\forall t_0 \geq T$,

$$\begin{aligned} & \int_{t_0}^{t_0+T_{\varepsilon,0}} \|q^\top \hat{\varphi}_{\varepsilon,b}(\tau)\|^2 d\tau \\ & \leq 2 \int_{t_0}^{t_0+T_{\varepsilon,0}} \|q^\top \varphi_\varepsilon(\tau)\|^2 d\tau + 2 \int_{t_0}^{t_0+T_{\varepsilon,0}} \|q^\top \vartheta_{\varepsilon,b}(\tau)\|^2 d\tau \\ & \leq 2c_2\varepsilon^2 + 2\varepsilon^3T_{\varepsilon,0}\varpi_4^2 =: \hat{\beta}_{\varepsilon,2}. \end{aligned}$$

On the other hand, since $(a+b)^2 \geq \frac{1}{2}a^2 - b^2$, by the same reasoning used above, one has that, for all $t_0 \geq T$,

$$\int_{t_0}^{t_0+T_{\varepsilon,0}} \|q^\top \hat{\varphi}_{\varepsilon,b}(\tau)\|^2 d\tau \geq \frac{1}{2}c_1\varepsilon^2 - \varepsilon^3T_{\varepsilon,0}\varpi_4^2 =: \hat{\beta}_{\varepsilon,1}.$$

Thus, if (16) holds, then there exists $\varepsilon^* > 0$ such that (17) holds for all $\varepsilon \in (0, \varepsilon^*)$, with $\check{c}_1 = \frac{1}{4}c_1$, $\check{c}_2 = 4c_2$, and $\hat{T}_{\varepsilon,0} = T_{\varepsilon,0}$. $\square$

Theorem 2. Let Assumption 1 hold, assume that ϕ is bounded and globally Lipschitz, and that the solution $x(t)$ of system (2) is bounded. Thus, consider system (13) with $\gamma = \varepsilon^{n-1}$ and $M \geq 2\|\lambda\|^2$. There exist $\varepsilon^* > 0$, $\Theta^* > 0$, and $\Delta^* > 0$ such that, if $\varepsilon < \varepsilon^*$, $\|\text{col}(\tilde{x}_b(0), \tilde{\lambda}_b(0))\|^2 \leq \Delta^*$, and the signal $\varphi_\varepsilon(t)$ satisfies (16) with $\frac{T_{\varepsilon,0}}{c_1} \leq \Theta^*$ for all $\varepsilon \in (0, \varepsilon^*)$, then

$$\lim_{t \rightarrow +\infty} \hat{x}_b(t) - x(t) = 0, \quad \lim_{t \rightarrow +\infty} \hat{\lambda}_b(t) - \lambda = 0. \quad (18)$$

Proof. By Theorem 1 and Lemma 2, for all $T > 0$, all $\Delta > 0$, and all $\varrho > 0$, there exists $\varepsilon^\diamond > 0$ such that if $\varepsilon < \varepsilon^\diamond$ and $\|\tilde{x}_{\varepsilon,b}(0)\|^2 \leq \Delta$, then $\|\tilde{x}_b(t)\|^2 \leq \varrho$ and (17) holds, for all $t \geq T$. Thus, noticing that the dynamics (13) matches those of system (5) for all times $t \geq 0$ such that $\|\hat{\lambda}_b(t)\|^2 < M$, there exist $\varepsilon^* \in (0, \varepsilon^\diamond)$ and $\Theta^* > 0$ such that, if $\varepsilon < \varepsilon^*$, the signal $\varphi_\varepsilon(t)$ satisfies (16) with $\frac{T_{\varepsilon,0}}{c_1} \leq \Theta^*$ for all $\varepsilon \in (0, \varepsilon^*)$, and $\|\tilde{x}_{\varepsilon,b}(0)\| \leq \Delta$, then, letting $\tilde{x}_{\varepsilon,b}$ and $\tilde{\lambda}_b$ be defined as in the proof of Theorem 1 and letting $\tilde{\xi}_{\varepsilon,b} := \text{col}(\tilde{x}_{\varepsilon,b}, \tilde{\lambda}_b)$, one has that $\|\tilde{\xi}_{\varepsilon,b}(t)\|^2 \leq \varsigma_1 e^{\varsigma_2(t-t_0)}\|\tilde{\xi}_{\varepsilon,b}(t_0)\|^2$, for all $t \geq t_0 \geq T$, where, as $\varepsilon \rightarrow 0$,

$$\varsigma_1 \leq \frac{\max\{1, \bar{\sigma}\}}{\min\{1, \underline{\sigma}\}} \quad \varsigma_2 \geq \frac{\varepsilon^{2n-1}\nu_a}{32 \max\{1, \bar{\sigma}\}^2 \Theta^*}, \quad (19)$$

ν_a is the positive scalar satisfying (8), $\bar{\sigma}$ and $\underline{\sigma}$ denote the minimum and maximum eigenvalue of the matrix P_a satisfying (8), provided that

$$\|\tilde{\xi}_{\varepsilon,b}(T)\|^2 < \frac{1}{\varsigma_1}(M - \|\lambda\|^2). \quad (20)$$

Therefore, it remains to prove that (20) holds. Since $\|\tilde{x}_b(T)\|^2 \leq \varrho$, one has that (20) holds if $\|\tilde{\lambda}_b(T)\|^2 < \frac{1}{\varsigma_1}(M - \|\lambda\|^2) - \varrho$. By noticing that $C_0\tilde{x}_b = C_0\tilde{x}_{\varepsilon,b}$ and that (15) holds, one has

$$\|\tilde{\lambda}_b(T)\|^2 \leq 2\|\tilde{\lambda}_b(0)\|^2 + 2\varepsilon^{2n-2}\bar{\phi}^2(\varpi_1 + \varepsilon^{2n-1}\varpi_2)T^2,$$

where $\bar{\phi} := \sup_{x \in \mathbb{R}^n} \phi(x)$. Therefore, since T and ϱ can be made arbitrarily small by letting ε be sufficiently small by Theorem 1, if $\|\tilde{\lambda}_b(0)\|$ is sufficiently small, then (20) holds, thus implying that (18) holds. $\square$

By Theorems 1 and 2, the adaptive observer (13) preserves the practical convergence properties of the classical high-gain observer (3) in the case that the signal $\varphi_\varepsilon(t)$ does not satisfy the persistence of excitation condition (16), whereas, if this condition holds, it asymptotically reconstructs both the state $x(t)$ and the parameter vector λ of system (2).

This section is concluded by discussing the properties of the adaptive observer (13) via several remarks.

Remark 1. In Theorem 2, it has been assumed that $\gamma = \varepsilon^{n-1}$ so to guarantee that the constant ς_1 in (19) is independent of ε . However, also other choices can be made to improve the convergence speed of the observer (13), although, in such cases, the bound on $\|\tilde{\lambda}_b(0)\|$ that ensures absence of saturation depends on γ and ε .

Remark 2. By the proof of Theorem 2, the bound Δ^* on the initial error can be made arbitrarily large by letting the saturation parameter M be sufficiently large.

Remark 3. By (15), the convergence rate of the error $\tilde{x}_b(t)$ to a neighborhood of the origin increases as the gain $\frac{1}{\varepsilon}$ of the observer (13) is increased, whereas, by (19), the convergence rate of the error $\text{col}(\tilde{x}_b, \tilde{\lambda}_b)$ decreases as the gain $\frac{1}{\varepsilon}$ increases. Therefore,

the gain $\frac{1}{\varepsilon}$ has to be selected sufficiently large to guarantee that $\varepsilon \leq \varepsilon^*$, but not too large to avoid slow convergence of $\hat{\lambda}_b(t)$ to 0.

Remark 4. If the function v is partially known, i.e., $v(x) = \lambda^\top \phi(x) + \phi_0(x)$, with ϕ and ϕ_0 being known functions, then the observer (13) can be modified as

$$\dot{\hat{x}}_b(t) = A_\varepsilon \hat{x}_b(t) + B_\varepsilon y(t) + H(\hat{\lambda}_b^\top(t) \phi(\hat{x}_b(t)) + \phi_0(\hat{x}_b(t))), \quad (21a)$$

$$\dot{\hat{\lambda}}_b(t) = \mathcal{P}_M(-\gamma \tilde{y}_b(t) \phi(\hat{x}_b(t))), \quad (21b)$$

in order to estimate just the parameters of interest.

Remark 5. The adaptive observers given in this section can be also used to deal with time-varying, controlled nonlinear systems of the form

$$\dot{z}(t) = f(t, z(t), u(t)), \quad y(t) = h(t, z(t), u(t)), \quad (22)$$

where $z(t) \in \mathbb{R}^\ell$ is the state, $u(t) \in \mathbb{R}^p$ is the control input, f and h are real analytic functions. If system (22) is n -strongly differentially observable according to Definition 4.2 of Gauthier and Kupka (2001) and the input $u(t)$ is C^n , then it can be immersed in the following canonical observability form

$$\dot{x}_i(t) = x_{i+1}(t), \quad i = 1, \dots, n-1, \quad (23a)$$

$$\dot{x}_n(t) = v\left(t, x(t), u(t), \frac{du(t)}{dt}, \dots, \frac{d^n u(t)}{dt^n}\right), \quad (23b)$$

$$y(t) = x_1(t), \quad (23c)$$

Thus, if $u_{e,n}(t) := \text{col}(u(t), \frac{d}{dt}u(t), \dots, \frac{d^n}{dt^n}u(t))$ is measurable and $v(t, x, u_{e,n}) = \lambda^\top \phi(t, x, u_{e,n})$ for some known feature vector ϕ and some unknown parameter vector λ , with $\phi(t, x, u_{e,n})$ being globally Lipschitz in x , i.e., there exists $L > 0$ such that $|\phi(t, x, u_{e,n}) - \phi(t, \hat{x}, u_{e,n})| \leq L \|x - \hat{x}\|$ for all $\hat{x}, x \in \mathbb{R}^n$, all $u_{e,n} \in \mathbb{R}^{n_p}$, and all $t \geq 0$, then the observers given in this section can be used to estimate the state of system (23) by substituting $\phi(\hat{x}(t))$ with $\phi(t, \hat{x}(t), u_{e,n}(t))$.

Remark 6. All throughout this section, in order to simplify the exposition, it has been assumed that the function ϕ is globally Lipschitz and bounded. However, in the case that ϕ is just locally Lipschitz, but the state $x(t)$ of system (2) lies in a bounded, known compact set $\mathcal{X} \subset \mathbb{R}^n$, the proposed adaptive observers can still be applied by saturating the feature vector $\phi(\hat{x}(t))$; see Astolfi et al. (2018) and Teel (2016) for further details.

Remark 7. If the output $y(t)$ of system (2) is affected by an additive, sufficiently small measurement noise $\delta(t)$, then the dynamics of the error $\tilde{x}_{e,b}$ are given by

$$\begin{aligned} \dot{\tilde{x}}_{e,b}(t) &= \frac{1}{\varepsilon} A_1 \tilde{x}_{e,b}(t) + E_\varepsilon H \tilde{\lambda}_b^\top(t) \phi(\hat{x}_b(t)) \\ &\quad + E_\varepsilon H \lambda^\top (\phi(\hat{x}_b(t)) - \phi(x(t))) + \frac{1}{\varepsilon} B_1 \delta(t), \end{aligned}$$

whereas the dynamics of the error $\tilde{\xi}_{e,b}$ are given, for all $t \geq 0$ such that $\|\hat{\lambda}_b(t)\|^2 < M$, by

$$\dot{\tilde{\xi}}_{e,b}(t) = \bar{A}_b(t) \tilde{\xi}_{e,b}(t) + \frac{1}{\varepsilon} \bar{B}_1 \delta(t),$$

with $\bar{A}_b(t)$ defined as in the proof of Theorem 2, $\bar{B}_1 := \text{col}(B_1, 0)$, and $B_1 := \text{col}(k_1, \dots, k_n)$. Therefore, by trivial extensions of the proofs of Theorems 1 and 2, even in the case that the amplitude of the disturbance δ and the observer gain $\frac{1}{\varepsilon}$ are sufficiently small to let the persistence of excitation hypothesis (17) be satisfied, the amplitude of the norm of the errors $\tilde{x}_b$ and $\tilde{\lambda}_b$ increases as the gain $\frac{1}{\varepsilon}$ is increased.

4. Linear-in-parameters neural adaptive observers for nonlinear systems

The main objective of this section is to show how to design adaptive observers for system (2) in the case that Assumption 1 does not hold. Toward this goal, assume that the state $x(t)$ of system (2) lies in a compact set $\mathcal{X} \subset \mathbb{R}^n$ for all $t \geq 0$ and let $\{\psi_i\}_{i=1}^\infty$ be a dense basis in the Sobolev norm $\mathcal{W}^{1,\infty}$ according to the definition given in Chapter 1 of Adams and Fournier (2003). For instance, the basis functions ψ_i can be chosen as *monomials* in x , as *recurrent higher order neural networks* (see Section 2 of Kosmatopoulos et al., 1997), i.e.,

$$\psi_i(x) = \prod_{j=1}^n \left(\frac{1}{1 + e^{-\alpha_{i,j} x_j}} + \beta_{i,j} \right)^{\gamma_{i,j}}, \quad (24)$$

with $\alpha_{i,j} \in \mathbb{R}$, $\beta_{i,j} \in \mathbb{R}$, and $\gamma_{i,j} \in \mathbb{Z}$, $\gamma_{i,j} > 0$, $j = 1, \dots, n$, or as *radial basis functions*, i.e.,

$$\psi_i(x) = e^{-\beta_i \|x - \alpha_i\|^2},$$

with $\alpha_i \in \mathbb{R}^n$ and $\beta_i > 0$. By the Stone–Weierstrass theorem (Theorem 7.32 of Rudin, 1976), for each $\omega > 0$ there exist $m \in \mathbb{Z}$, $m > 0$, and $\lambda^* \in \mathbb{R}^m$ such that

$$|v(x) - (\lambda^*)^\top \psi(x)| \leq \omega, \quad \forall x \in \mathcal{X}, \quad (25)$$

where $\psi(x) = [\psi_1(x) \ \dots \ \psi_m(x)]^\top$. Consider the neural adaptive observer with state $(\hat{x}_n(t), \hat{\lambda}_n(t)) \in \mathbb{R}^n \times \mathbb{R}^m$,

$$\dot{\hat{x}}_n(t) = A_\varepsilon \hat{x}_n(t) + B_\varepsilon y(t) + H \hat{\lambda}_n^\top(t) \psi(\hat{x}_n(t)), \quad (26a)$$

$$\dot{\hat{\lambda}}_n(t) = \mathcal{P}_M(-\gamma \tilde{y}_n(t) \psi(\hat{x}_n(t))), \quad (26b)$$

where $\tilde{y}_n := \hat{y}_n - y$ and $\hat{y}_n = C_0 \hat{x}_n$.

Theorem 3. Suppose that ψ is globally Lipschitz and that $x(t) \in \mathcal{X}$ for all $t \geq 0$. Consider system (26) and let $\|\hat{\lambda}_n(0)\|^2 \leq M$. For each $\varrho > 0$, $\Delta > 0$, and $T > 0$, there exists $\varepsilon^* > 0$ such that, if $\varepsilon < \varepsilon^*$ and $\|\hat{x}_n(0) - x(0)\|^2 \leq \Delta$, then $\hat{x}_n(t)$ and $\hat{\lambda}_n(t)$ are bounded and

$$\|\hat{x}_n(t) - x(t)\|^2 \leq \varrho, \quad \forall t \geq T. \quad (27)$$

Proof. By the same reasoning used in the proof of Lemma 1, there exists a function $Z(x, \hat{x})$, $\|Z(x, \hat{x})\| \leq L$, $\forall x, \hat{x} \in \mathbb{R}^n$, such that $\psi(\hat{x}) - \psi(x) = Z(x, \hat{x})\tilde{x}$. Hence, consider the errors $\tilde{x}_n := \hat{x}_n - x$, $\tilde{x}_{e,n} := E_\varepsilon \tilde{x}_n$, where E_ε is defined as in (7), and $\tilde{\lambda}_n := \hat{\lambda}_n - \lambda^*$. The dynamics of the error $\tilde{x}_{e,n}$ are given by

$$\begin{aligned} \dot{\tilde{x}}_{e,n}(t) &= \left(\frac{1}{\varepsilon} A_1 + E_\varepsilon H (\lambda^*)^\top Z(x(t), \hat{x}_n(t)) E_\varepsilon^{-1} \right) \tilde{x}_{e,n} \\ &\quad + \varepsilon^{n-1} H \psi^\top(\hat{x}_n(t)) \tilde{\lambda}_n(t) + \varepsilon^{n-1} H \theta_n(t), \end{aligned}$$

where $\theta_n(t) := (\lambda^*)^\top \psi(x(t)) - v(x(t))$. Hence, letting P be the solution to the Lyapunov equation (14) and letting q_b be defined as in the proof of Theorem 1, consider the function $V_n(t) = \tilde{x}_{e,n}^\top(t) P \tilde{x}_{e,n}(t)$, whose time derivative satisfies

$$\begin{aligned} \dot{V}_n(t) &\leq -\left(\frac{1}{\varepsilon} - 2L \|q_b\| \|\lambda\| - 2\|q_b\|^2 \right) \|\tilde{x}_{e,n}(t)\|^2 \\ &\quad + \varepsilon^{2n-2} (\tilde{\lambda}_n^\top(t) \phi(\hat{x}_n(t)))^2 + \varepsilon^{2n-2} \theta_n^2(t). \end{aligned}$$

Thus, by the same reasoning used to prove Theorem 1 and since $|\theta_n(t)| \leq \omega$ for all $t \geq 0$, one has that there exists $\varepsilon^\diamond > 0$ such that, if $\varepsilon < \varepsilon^\diamond$ and

$$\|\tilde{x}_{e,n}\|^2 > \frac{\varepsilon^{2n-1} (2\bar{M}\bar{\phi}^2 + \omega^2)}{1 - 2\varepsilon L \|q_b\| \|\lambda\| - 2\varepsilon \|q_b\|^2 - 2\varepsilon \bar{M} L^2},$$

then $\dot{V} < 0$. Therefore, there exist constants $v_1 > 0$ and $v_2 > 0$, independent of ε , such that, for all $t \geq t_0 \geq 0$,

$$\|\tilde{x}_{e,n}(t)\|^2 \leq v_1 e^{-\frac{1}{2\varepsilon}(t-t_0)} \|\tilde{x}_{e,n}(t_0)\|^2 + \varepsilon^{2n-1} v_2.$$

Therefore, the statement follows by the same reasoning used to prove [Theorem 1](#). $\square$

Theorem 4. Assume that ψ is bounded and globally Lipschitz, and that $x(t) \in \mathcal{X}$ for all $t \geq 0$. Thus, consider system (26) with $\gamma = \varepsilon^{n-1}$ and $M \geq 2 \|\lambda^*\|^2$. There exist $\varepsilon^* > 0$, $\Theta^* > 0$, and $\Delta^* > 0$ such that, if $\varepsilon < \varepsilon^*$, $\|\text{col}(\tilde{x}_n(0), \tilde{\lambda}_n(0))\| \leq \Delta^*$, the signal $\psi_\varepsilon(t) := ((sI - \frac{1}{\varepsilon} A_1) H \psi(x(t)))^\top$ satisfies

$$c_1 \varepsilon^2 I \leq \int_{t_0}^{t_0+T_{\varepsilon,0}} \psi_\varepsilon(\tau) \psi_\varepsilon^\top(\tau) d\tau \leq c_2 \varepsilon^2 I,$$

for all $t_0 \geq 0$, for some constants $c_1 > 0$ and $c_2 > 0$ independent of ε , with $\frac{T_{\varepsilon,0}}{c_1} \leq \Theta^*$ for all $\varepsilon \in (0, \varepsilon^*)$, then, for each $\rho > 0$, there exists $\omega_\varepsilon^* > 0$ (dependent on the gain $\frac{1}{\varepsilon}$) such that, if (25) holds with $\omega \leq \omega_\varepsilon^*$, then

$$\limsup_{t \rightarrow +\infty} \|\hat{x}_n(t) - x(t)\| \leq \rho, \quad \limsup_{t \rightarrow +\infty} \|\hat{\lambda}_n(t) - \lambda^*\| \leq \rho.$$

Proof. Let $\tilde{x}_n$, $\tilde{\varepsilon}_{\varepsilon,n}$, and $\tilde{\lambda}_n$ be defined as in the proof of [Theorem 3](#) and let $\tilde{\xi}_{\varepsilon,n} := \text{col}(\tilde{x}_{\varepsilon,n}, \tilde{\lambda}_n)$, whose dynamics are given, for all $t \geq 0$ such that $\|\hat{\lambda}_n(t)\|^2 < M$, by

$$\dot{\tilde{\xi}}_{\varepsilon,n}(t) = \bar{A}_n(t) \tilde{\xi}_{\varepsilon,n}(t) + \bar{F}_n \theta_n(t), \quad (28)$$

with $\theta_n(t)$ defined as in the proof of [Theorem 3](#) and

$$\bar{A}_n(t) := \begin{bmatrix} \frac{1}{\varepsilon} A_1 + E_\varepsilon H(\lambda^*)^\top Z(x(t), \hat{x}_n(t)) E_\varepsilon^{-1} E_\varepsilon H \psi^\top(\hat{x}_n(t)) \\ -\gamma \psi(\hat{x}_n(t)) C_0 & 0 \end{bmatrix},$$

$$\bar{F}_n(t) := \begin{bmatrix} \varepsilon^{n-1} H \\ 0 \end{bmatrix}.$$

By [Theorem 3](#) and a reasoning wholly similar to the one used to prove [Lemma 2](#), for all $T > 0$, all $\Delta > 0$, and all $\varrho > 0$, there exists $\varepsilon^\circ > 0$ such that if $\varepsilon < \varepsilon^\circ$ and $\|\tilde{x}_{\varepsilon,n}(t_0)\|^2 \leq \Delta$, then $\|\tilde{x}_n(t)\|^2 \leq \varrho$ for all $t \geq T$ and the signal $\hat{\psi}_{\varepsilon,n}(t) := ((sI - \frac{1}{\varepsilon} A_1) H \psi(\hat{x}_n(t)))^\top$ satisfies

$$\check{c}_1 \varepsilon^2 I \leq \int_{t_0}^{t_0+\hat{T}_{\varepsilon,0}} \hat{\psi}_{\varepsilon,n}(\tau) \hat{\psi}_{\varepsilon,n}^\top(\tau) d\tau \leq \check{c}_2 \varepsilon^2 I, \quad (29)$$

for all $t_0 \geq T$ and for some $\check{c}_1 > 0$, $\check{c}_2 > 0$ independent of ε . Thus, by the reasoning used in the proofs of [Lemma 1](#) and of [Theorem 2](#), and using the bound given in Lemma 3.3.3 of [Ioannou and Sun \(2012\)](#), if $\varepsilon \leq \varepsilon^*$ and $\frac{T_{\varepsilon,0}}{c_1} \leq \Theta^*$ for all $\varepsilon \in (0, \varepsilon^*)$, then one has that, for all $t \geq T$ such that $\|\hat{\lambda}_n(\tau)\|^2 < M$ for all $\tau \in [T, t]$, $\|\tilde{\xi}_{\varepsilon,n}(t)\| \leq \left(\frac{\bar{\sigma}}{\underline{\sigma}}\right)^{\frac{1}{2}} e^{-\frac{\varepsilon^2 n-1}{64 \bar{\sigma}^2 \Theta^*} (t-T)} \|\tilde{\xi}_{\varepsilon,n}(T)\| + \frac{64 \bar{\sigma}^{\frac{5}{2}} \Theta^*}{\underline{\sigma}^{\frac{1}{2}} v_a} \frac{\omega}{\varepsilon^n}$,

where $\bar{\sigma}$ and $\underline{\sigma}$ are the minimum and maximum eigenvalue of P_a in (8). Hence, the statement follows by the same reasoning used to prove [Theorem 2](#). $\square$

By [Theorem 4](#), the mismatch error θ_n acts on the dynamics of the error $\tilde{\xi}_{\varepsilon,n}$ as a disturbance, whose effects are amplified by the gain $\frac{1}{\varepsilon}$. Therefore, as already pointed out in [Remarks 3](#) and [7](#), the gain $\frac{1}{\varepsilon}$ of the neural adaptive observer (26) has to be selected sufficiently large to let $\varepsilon < \varepsilon^*$, but not too large to avoid decreasing the convergence speed and amplifying the effects of the mismatch error θ_n and of eventual measurements noise. Furthermore, as highlighted in [Remark 1](#) for system (13), one can make other choices than ε^{n-1} for the adaptation gain γ in order to improve the convergence speed of the neural adaptive observer (26). Finally, note that by the same reasoning given in [Remark 5](#), the neural adaptive observer (26) can be also used to estimate the state of controlled nonlinear systems, provided that the input and its time derivatives lie in a compact set.

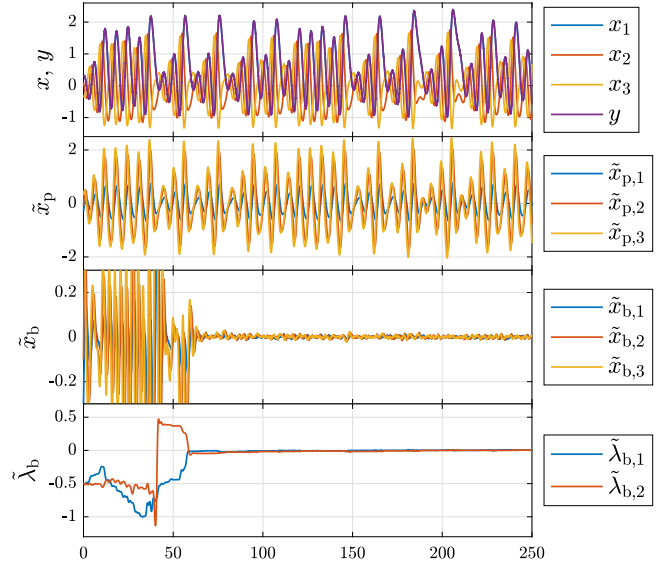


Fig. 1. Time behavior of the state x of system (31) and of the estimation errors $\tilde{x}_p = \hat{x}_p - x$, $\tilde{x}_b = \hat{x}_b - x$, and $\tilde{\lambda}_b = \hat{\lambda}_b - \lambda$ obtained using the practical high-gain observer (3) and the adaptive observer (21), respectively, in [Example 1](#) assuming that the output is affected by a Gaussian measurement noise.

5. Numerical examples

In this section, the effectiveness of the adaptive observers (13) and (26) is validated via simulations.

Example 1. Consider the Rössler prototype-4 system given in Equation (3.4) of [Sprott \(2010\)](#)

$$\dot{z}_1(t) = -z_2(t) - z_3(t), \quad (30a)$$

$$\dot{z}_2(t) = z_1(t), \quad (30b)$$

$$\dot{z}_3(t) = \lambda_1(z_2(t) - z_2^2(t)) - \lambda_2 z_3(t), \quad (30c)$$

$$y(t) = z_2(t). \quad (30d)$$

By computing the observability map of order 3 of system (30) and letting $x = O_3(z)$, system (30) can be recast into the observability canonical form

$$\dot{x}_1(t) = x_2(t), \quad (31a)$$

$$\dot{x}_2(t) = x_3(t), \quad (31b)$$

$$\dot{x}_3(t) = [\lambda_1 \quad \lambda_2] \phi(x) + \phi_0(x), \quad (31c)$$

$$y(t) = x_1(t). \quad (31d)$$

where $\phi(x) = [-x_1 + x_1^2 \quad -x_1 - x_3]^\top$ and $\phi_0(x) = -x_2$. Letting $m = 2$, the adaptive observer (21) has been used to estimate the state of system (31) from noisy measurements of the output $y(t)$. Namely, in order to evaluate the effectiveness of such an observer in the presence of measurement noise, a numerical simulation has been carried out letting $\varepsilon = 0.9$, $k_1 = 2$, $k_2 = 2$, $k_3 = 1$, $\gamma = 1$, $M = 1$, $\lambda_1 = 0.5$, $\lambda_2 = 0.5$, $z(0) = [0.1 \quad 0.3 \quad 0]^\top$, $x(0) = O_3(z(0))$, $\hat{x}_b(0) = [0 \quad 0 \quad 0]^\top$, $\hat{\lambda}_b(0) = [0 \quad 0]^\top$, and adding band-limited, Gaussian, white noise to $y(t)$, with zero mean, sampling time equal to 0.1, and standard deviation equal to 0.01. For comparison, also the time behavior of the estimate $\hat{x}_p$ obtained using the practical high-gain observer (3) with the same gain ε and the same values k_1 , k_2 , k_3 has been simulated. [Fig. 1](#) depicts the results of such a simulation. As shown by such a figure, the adaptive high-gain observer (21) has improved performance with respect to the plain one (3) thanks to the fact

that it also estimates the function v governing the dynamics of system (2).

Another numerical simulation has been carried out in the same setting to compare the performance of the adaptive observer (21) with that of the extended Kalman filter (EKF) given in Eqs. (3) and (4) of [Bonnabel and Slotine \(2014\)](#), with $Q = I$, $R = 1$, and $P_x(0) = I$, and the unscented Kalman observer (UKO) given in Equation (16) of [Daid et al. \(2021\)](#), with $c = 1$, $\beta = 2$, $\lambda = -2$, and $P(0) = I$, both designed to estimate the state of system (31), where the parameters λ_1 and λ_2 have been considered as additional state variables with null time derivative. The results of these simulations are shown in Fig. 2.

First, note that the dynamics of the adaptive observer (21) are simpler than those of the EKF and UKO. In fact, system (21) has $n + m$ states, whereas the EKF and the UKO have $n + m + (n + m)^2$ states. Furthermore, by comparing Figs. 1 and 2, the sensitivity to Gaussian measurement noise of the estimate obtained using the adaptive observer (21) is comparable to that obtained using the EKF and the UKO, although the convergence time of the former is larger.

To further compare the EKF, the UKO, and the adaptive observer (21), Monte Carlo simulations have been carried out with the same parameters of above, but with increasing values of the standard deviation of the noise added to the output y . Fig. 3 depicts the mean estimation errors gathered at steady-state in such simulations.

As shown by such a figure, the performance of the adaptive observer (21) in the presence of measurement noise is similar to that of the EKF and the UKO.

Example 2. Consider the following model of a well-mixed, constant-volume, isothermal batch reactor (see Section 3.2 of [Haseltine & Rawlings, 2005](#))

$$\dot{z}_1 = \frac{1}{20}z_2z_3 - \frac{1}{2}z_1 \quad (32a)$$

$$\dot{z}_2 = -\frac{2}{5}z_2^2 - \frac{1}{20}z_3z_2 + \frac{1}{2}z_1 + \frac{1}{50}z_3 \quad (32b)$$

$$\dot{z}_3 = \frac{1}{5}z_2^2 - \frac{1}{20}z_2z_3 + \frac{1}{2}z_1 - \frac{1}{100}z_3, \quad (32c)$$

$$y = \frac{821}{25}z_1 + \frac{821}{25}z_2 + \frac{821}{25}z_3. \quad (32d)$$

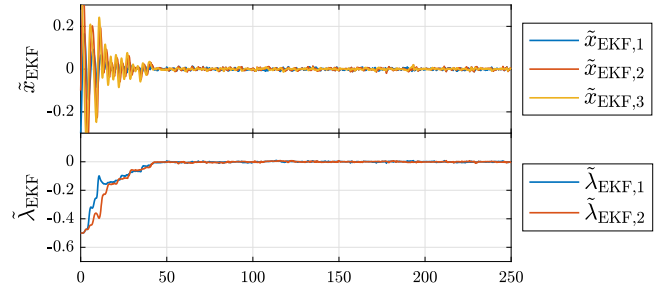
As shown in [Haseltine and Rawlings \(2005\)](#) and confirmed by a simulation whose results are shown in Fig. 4, both the EKF (with reference to Eqs. (3), (4) of [Bonnabel & Slotine, 2014](#), its parameters have been chosen as $Q = I$, $R = 1$, $P_z(0) = 10I$, and $\hat{z}(0) = [0 \ 0 \ 4]^T$) and the UKO (with reference to Equation (16) of [Daid et al., 2021](#), its parameters have been chosen as $c = 1$, $\beta = 2$, $\lambda = -2$, $P(0) = 10I$, and $\hat{z}(0) = [0 \ 0 \ 4]^T$) fail to estimate the state of system (32) with $z(0) = [0.5 \ 0.05 \ 0]^T$ due to the presence of multiple optima and to the fact that the initial guess is far from the actual initial condition of the system.

Assuming that the model (32) is known but the function v in (2) is unknown, the neural adaptive observer (26), with $m = 4$ and ψ being as in (24) with randomly picked weights, has been used to obtain estimates $\hat{x}$ of $x = O_3(z)$ and the dynamical inversion algorithm

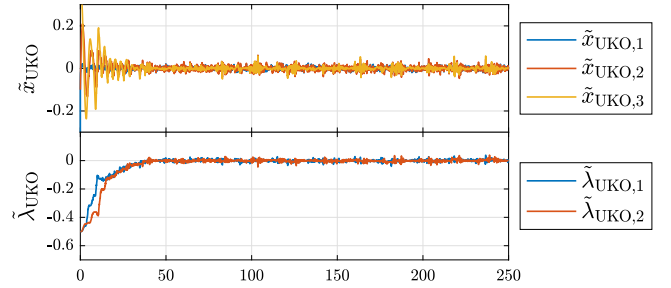
$$\dot{\hat{z}}_n = f(\hat{z}_n) + \mu \left(\frac{\partial O_3}{\partial z}(\hat{z}_n) \right)^T (\hat{x}_n - O_3(\hat{z}_n)), \quad (33)$$

given in [Menini et al. \(2017b\)](#) has been used to gather estimates $\hat{z}_n$ of the state of system (32) from $\hat{x}$. A numerical simulation has been carried out to test such a state observer letting $\varepsilon = 0.3$, $\gamma = 20$, $M = 10$, $\mu = 10$, $\hat{z}_n(0) = [0 \ 0 \ 4]^T$, and choosing $\hat{x}(0)$ at random. The results of this simulation are shown in Fig. 5

As shown by Fig. 5, in the case that the function v is not known in parametric form, the neural adaptive observer (26) practically estimates $x = O_3(z)$. Further, assuming that the dynamics (32)



(a) Estimation errors gathered using the EKF.



(b) Estimation errors gathered using the UKO.

Fig. 2. Time behavior of the estimation errors obtained using the EKF and the UKO in [Example 1](#) assuming that the output y is affected by a Gaussian measurement noise.

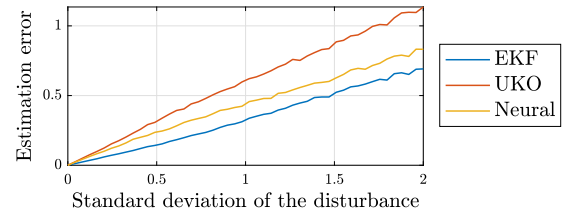


Fig. 3. Results of Monte Carlo simulations for the sensitivity of the EKF, the UKO, and the observer (21) to measurement noise in [Example 1](#).

are known (so to compute the observability map O_3), coupling this observer with the dynamical inversion algorithm (33) one can also estimate the state of system (32).

In order to further compare the neural adaptive observer (26) with the EKF and the UKO, numerical simulations have been carried out by initializing the EKF, the UKO, and the neural adaptive observer (26) at the same $\hat{z}(0) = z(0) + \zeta[0 \ 0 \ 4]^T$, with $\zeta \in [0, 1]$, and the performance of the three observers have been compared in terms of the cumulative error $\int_0^\infty (\hat{z}(\tau) - z(\tau))^T (\hat{z}(\tau) - z(\tau)) d\tau$. Fig. 6 depicts the results of such simulations.

As shown by such a figure, the performance of the EKF and of the UKO are quite similar and the estimation error obtained using these two observers is small when ζ is small, i.e., when the initial guess is close to the actual initial condition of the system. On the other hand, the performance of the neural adaptive observer (26) is worse than that of the EKF and the UKO for small values of ζ , and better for large values of ζ . In particular, note that for $\zeta > 0.3316$, neither the EKF nor the UKO converge to the actual state of system (32).

6. Conclusions

Neural adaptive observers for nonlinear systems in observability canonical form have been proposed. In the case that a parametric model for the observed plant is available, it has been

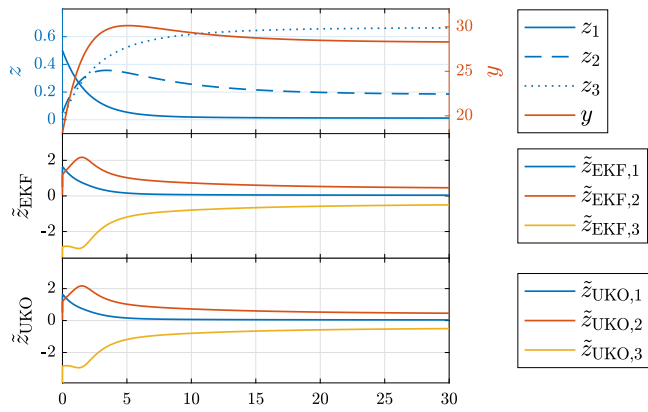


Fig. 4. Estimation errors obtained using the EKF and the UKO to estimate the state of system (32) in Example 2.

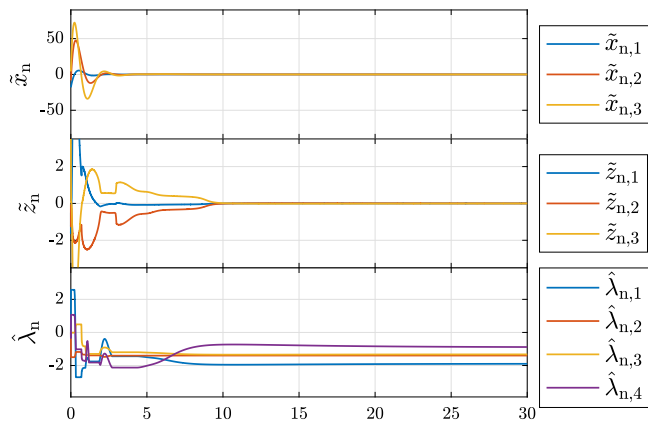


Fig. 5. Estimation errors obtained using the observer (26) and the dynamical inversion algorithm (33) in Example 2.

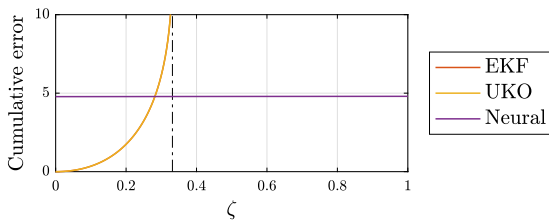


Fig. 6. Performance of the EKF, the UKO, and the neural adaptive observer (26) for different initial conditions.

shown that, under a persistence of excitation condition on the feature vector, these observers are capable of asymptotically reconstructing both the state and the unknown parameters of the observed plant. Even in the case that such a hypothesis does not hold, the proposed neural adaptive observers still provide practical estimates of both the state and the model of the observed plant. The effectiveness of these observers has been validated via numerical simulations.

Differently from the observers given in Kim et al. (1996), the adaptive observers (13) and (26) do not rely on any strictly positive real assumption, whereas, differently from Abdollahi et al. (2006), Adhyaru (2012), Chakrabarty et al. (2019), Hovakimyan et al. (2002), Lakhal et al. (2010), Leu et al. (1999) and Vargas and Hemerly (2000), they exploit a high-gain structure to guarantee that the estimation error becomes arbitrarily small arbitrarily fast by tuning the single parameter ε even in the case that the persistence of excitation condition (16) does not hold. Finally, differently from the observers given in Bin and Marconi (2021)

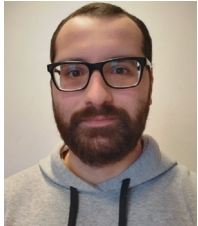
and Buisson-Fenet et al. (2021), the adaptation parameters are updated continuously in time rather than via iterative procedures.

As the neural observer given in Kim et al. (1996), the ones proposed in this paper are linear-in-parameters. Future work will deal with the extension of the results given in this paper to the nonlinear-in-parameters case, thus possibly allowing to take into account models having higher degrees of nonlinearity.

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